Healthcare Risk Prediction using Machine Learning and Neural Networks

Assignment Report

Dataset	healthcare_dataset.csv
Total Records	6,000 patients
Target Variable	diabetes (0 = No, 1 = Yes)
Models Evaluated	Logistic Regression and Neural Network

1. Problem Understanding

The objective of this project is to develop predictive models that estimate whether a patient is at risk of diabetes by using routine health indicators such as age, body mass index (BMI), weekly exercise level, smoking status, and blood pressure. The task is a binary classification problem in which the model must assign each patient to either the diabetic or non-diabetic category.

Predicting diabetes risk is important because the disease often develops gradually and may remain undetected until complications begin to appear. A reliable prediction system can support earlier screening, guide preventive care, and help healthcare providers prioritize patients who may benefit from lifestyle counselling, monitoring, or further clinical testing.

Why this problem matters in practice:

- Early identification can reduce the risk of long-term complications such as kidney damage, cardiovascular disease, and nerve problems.
- Risk prediction supports better allocation of healthcare resources by helping providers focus attention on high-risk individuals.
- Preventive action is usually less costly and more effective than late-stage treatment.

2. Data Exploration Insights

The dataset contains 6,000 patient records and 7 columns, including one identifier field and five health-related predictor variables. No missing values were found in any column, which means the data was ready for modeling without additional imputation. The target variable is imbalanced: 5,458 patients (90.97%) are non-diabetic and 542 patients (9.03%) are diabetic. This imbalance is important because it can make model accuracy look strong even when the minority diabetic class is not being detected properly.

Key observations from the dataset:

- Patient age ranges from 18 to 84 years, with an average age of 51.11 years.
- The average BMI is 26.00, which indicates that a large part of the sample falls near or above the overweight range.
- The mean blood pressure is 134.24 mmHg, which is close to a clinically elevated level.
- Average weekly exercise is 3.01 hours, and lower activity appears more common in diabetic cases.

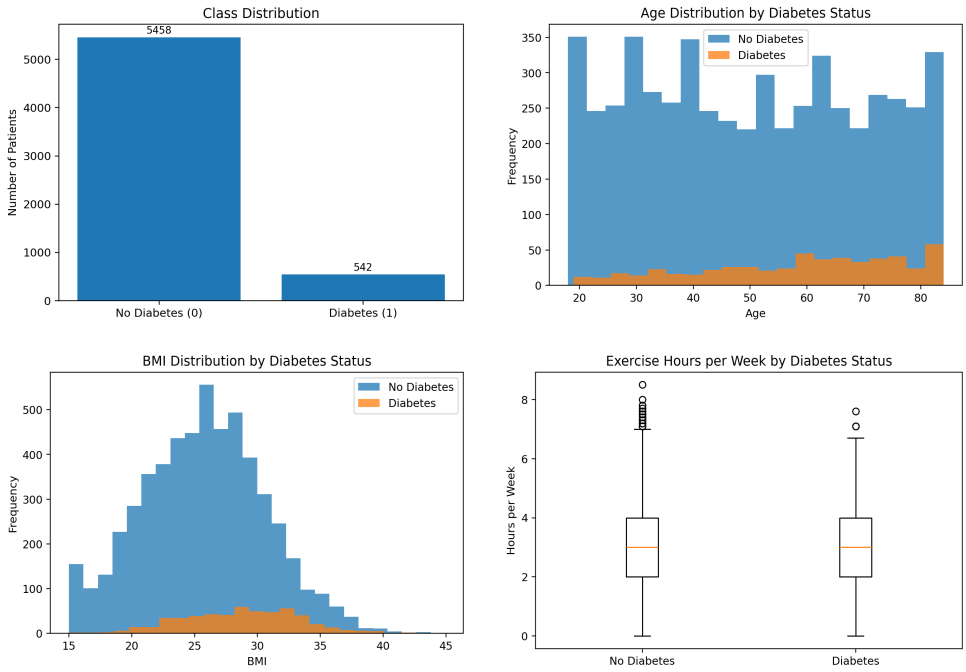

Figure 1. Automatically generated visualizations from the uploaded dataset: class distribution, age distribution, BMI distribution, and exercise pattern by diabetes status.

Important features influencing prediction

Correlation analysis shows that BMI has the strongest relationship with diabetes in this dataset, followed by blood pressure and age. Logistic Regression coefficients also support the same pattern after feature scaling, which means those variables contribute more strongly to the model's decision boundary than smoker status or exercise hours.

Feature	Correlation with Diabetes	Logistic Coefficient
Bmi	0.171	0.524
Blood Pressure	0.149	0.273
Age	0.119	0.337
Exercise Hours Per Week	-0.001	0.033
Smoker	-0.007	-0.176

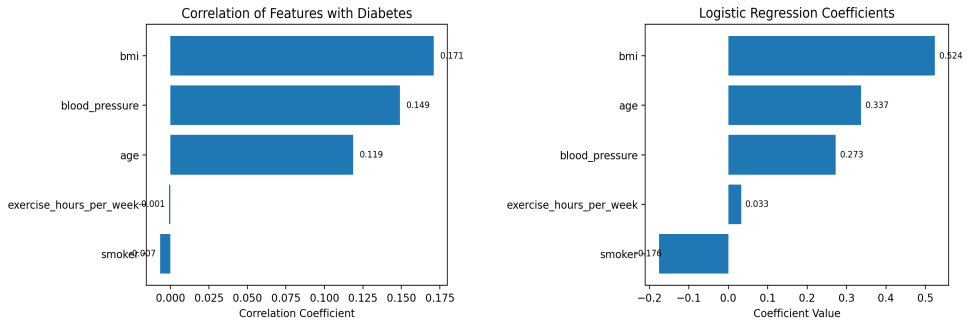

Figure 2. Feature relevance views generated from the uploaded dataset: correlations with the target variable and Logistic Regression coefficients.

3. Model Implementation

Two different approaches were implemented to solve the same classification problem so that their behavior could be compared on the same test set.

Machine Learning model - Logistic Regression

- The identifier column was removed because it does not provide predictive value.
- The remaining features were separated into predictors and target, then the dataset was split into training and testing subsets using stratified sampling.
- StandardScaler was applied so that variables measured on different scales, such as age and blood pressure, would contribute more consistently.
- Logistic Regression was trained with class_weight='balanced' to reduce the effect of class imbalance and improve detection of diabetic cases.

Neural Network architecture

- The neural network was built as a Sequential binary-classification model.
- Input: 5 features - age, BMI, exercise hours per week, smoker status, and blood pressure.
- Hidden layers: dense layers with ReLU activation were used to learn non-linear patterns.
- Output layer: one neuron with sigmoid activation to produce the probability of diabetes.
- Training used binary cross-entropy loss with an adaptive optimizer and early stopping to control overfitting.

4. Model Evaluation

Model performance was evaluated using the values obtained from the final test runs shared for both models. For the machine learning model, multiple classification metrics were considered because accuracy alone is not enough in an imbalanced healthcare dataset. For the neural network, overall test accuracy and loss were reported, and the confusion matrix was used to understand how predictions were distributed across classes.

Model	Accuracy	Precision	Recall	F1 Score	Loss
Logistic Regression	0.65	0.1518	0.6296	0.2446	-
Neural Network	0.91	-	-	-	0.2837

Interpretation of results

- Logistic Regression achieved lower overall accuracy, but it detected a meaningful portion of diabetic patients, giving it a higher practical value for screening.
- Its recall of 0.6296 indicates that the model correctly identified around 63% of the diabetic patients in the test set.
- Precision remained low because the model produced many false positives, which is visible in the confusion matrix.
- The Neural Network reached 91% accuracy, but the confusion matrix shows that it predicted almost every patient as non-diabetic. This means the high accuracy is largely a consequence of class imbalance rather than better clinical usefulness.

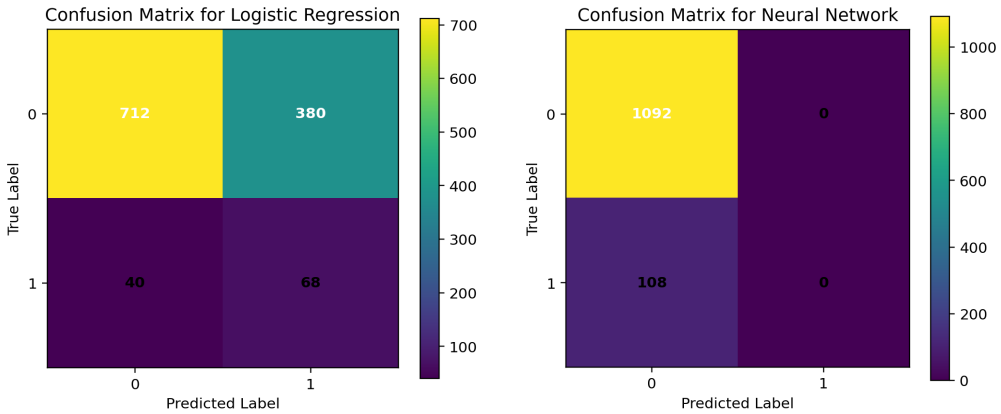

Figure 3. Confusion matrices based on the reported test results for Logistic Regression and the Neural Network.

Logistic Regression Confusion Matrix	Predicted 0	Predicted 1
Actual 0	712	380
Actual 1	40	68

5. Comparative Analysis

The comparison between the two models highlights why performance metrics must be interpreted carefully in medical prediction tasks. A model with high overall accuracy is not automatically the better model if it fails to detect the actual positive cases that matter most.

Differences between the models

- Logistic Regression is simpler, easier to interpret, and makes it possible to understand which variables push predictions toward diabetes risk.
- The Neural Network is more flexible and can model non-linear relationships, but in this case it appears to have converged toward predicting the majority class.
- Logistic Regression is weaker in precision, while the Neural Network is weaker in minority-class detection.

Reasons for the accuracy difference

- The dataset is imbalanced: about 90.97% of patients belong to the non-diabetic class.
- Because of this imbalance, a model can obtain high accuracy simply by predicting the majority class most of the time.
- The Neural Network's reported results show exactly this behavior: high accuracy with ineffective detection of diabetic patients.
- Logistic Regression, especially with class balancing, produces a more useful decision boundary for identifying positives, even though that lowers overall accuracy.

Recommendations for real-world use

- For healthcare screening, recall should be prioritized along with F1 score, because missing a diabetic patient is riskier than raising a false alarm.
- Before deployment, both models should be improved using techniques such as SMOTE, class weights, threshold tuning, or ensemble methods.
- At the current stage, Logistic Regression is the more suitable baseline because it is interpretable and detects diabetic cases more reliably than the Neural Network.
- Any production-ready system should be validated with additional clinical data and evaluated for fairness, calibration, and stability across patient groups.

Overall, this comparison shows that model selection in healthcare should be guided by clinical usefulness rather than by a single summary metric. In this dataset, the Logistic Regression model is more meaningful for diabetes risk screening, while the Neural Network needs further tuning before it can be considered for practical use.